

Smart Air-Quality Monitoring using Low-cost Sensors

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Abstract

This project's main aim is to develop a cost-effective air quality monitoring system using low-cost environmental sensors. The circuit system in this project is designed to collect real-time data on some of air quality parameters such as particulate matter mainly PM_{2.5}, harmful gases like CO, NO₂, NH₃ along with pressure, temperature. A microcontroller-based circuit integrates these sensors and transmits the collected data wirelessly through LoRa Module and updates value based on geospatial location. To enhance accessibility and user interaction, a mobile application has been developed, which visualizes sensor readings in a clear and user-friendly interface. The app not only shows live values but can also store past readings for analysis. This integration of hardware and software enables users to monitor air quality conveniently and take informed actions based on the data. The aim is to offer an affordable solution for real-time environmental monitoring, especially in areas lacking access to sophisticated equipment. Our approach showcases how affordable technology can be a powerful tool in raising environmental awareness and encouraging more sustainable lifestyles in urban as well as rural communities

Keywords: Air quality monitoring system, Low-cost sensors, Real-time data acquisition, PM_{2.5}, Harmful gases (CO, NO₂, NH₃), Microcontroller, LoRa communication, Geospatial data, Mobile application, Environmental awareness, Sustainable technology

1. Introduction

Air pollution is a major environmental issue and public health crisis, contributing to a significant number of unreasonable deaths each year, particularly from diseases such as lung cancer and heart disease. Effective air quality monitoring is essential to prevent health problems related to pollution, however, most of present monitoring systems fail to do proper monitoring due to lack of proper coverage. (Air Pollution and Health by M. L. V. et al., 2021). Air quality plays a crucial role in determining human health, environmental integrity and climate change. Poor air quality contributes to degradation of ecosystems and accelerates global warming, affecting biodiversity. (WHO: Air Quality and control: 2021) Traditional air quality monitoring methods are often costly due to high initial capital investment and maintenance requirements in their life time. These systems demand highly sophisticated equipment with regular calibration, and operational support, limiting their widespread use and making them inaccessible for many regions. (Pappas, A. et al. (2018)). Traditional air quality monitoring stations are expensive to install and maintain, leading to small deployment and limited spatial resolution in measurements. This limitation obstructs the ability to capture localized pollution variations effectively. (M. M. J., Air Quality Monitoring (2020) Traditional Air-Quality Sensors are commonly able to detect major air pollutants such as carbon monoxide (CO), particulate matter (PM), ozone (O₃) and nitrogen dioxide (NO₂) etc. These measurements helps in assessing air quality and provide critical data for environmental monitoring and health studies. (Veselka et al., 2019).

Low-cost air quality sensors behave as a perfect affordable alternatives, enabling more micro-scalar environmental monitoring. These sensors facilitate real-time data collection, making air quality information more accessible to communities. (Bertoldi, P. et al. (2017)). Information gathered from affordable air quality sensors can play an important role in spreading information about pollutant levels and their impacts among the public people. Such data also supports evidence-based decisions, helping local authorities to think more about responsive environmental policies. (Sharma et al., 2021). With the continuous and tremendous progress in sensor technology, the reduction in their size helps to integrate these sensors with android based devices such as mobiles etc. This integration empowers individuals to track their exposure to pollutants and evaluate their potential health risks in real-time, offering an approach to monitor their environment and well-being.

Therefore, The aim of our present project is to develop a small-scale low-cost sensor system that analyzes some of air-quality parameters like Carbon Monoxide (CO), Nitrogen Dioxide (NO₂), Ammonia (NH₃) and PM_{2.5} concentrations along with Temperature and Humidity. This sensor system also have a connection with an app developed with flutter which displays real time data and graphs to show trends over time. In addition to real-time monitoring, the app also stores past data of readings taken by sensors. The app can show suggestions based on pollutant levels, if they exceed, it shows precautionary measures. This feature helps individuals to know about real-time quality of air around them. With this information, they can better understand their environment and make smarter choices for their health.

2. Methodology

This section describes the methodology including description of sensors used, Sensors connection and circuit assembly, integration of sensors with LoRa Module and GPS module, development of a mobile application, Integration of circuit with an application using Firebase, Testing and Validation of the prototype, etc.

2.1. Description of sensors and equipments used

MICS-6814 is a low-power MEMS-based gas sensor designed to monitor Carbon Monoxide (CO), Nitrogen Dioxide(NO₂) and Ammonia(NH₃) in the atmosphere. The PM2.5 GP2Y1010AU0F is a sensor used to detect PM2.5 and larger airborne particles using an infrared light-emitting diode and photodetector. The BMP280 is a sensor which is used to measure atmospheric pressure and temperature, which is ideal for weather forecasting and altitude detection in IoT applications. The ESP-32 is a low-power dual-core microcontroller with built-in Wi-Fi and Bluetooth capabilities that makes it suitable for use in IoT projects requiring wireless communication. The NEO 6M UBLOX GPS module is a GPS receiver that provides real-time location data for navigation and update of location-based data. The LoRa module is a long-range wireless communication device designed to enable transmission over kilometers with minimal energy consumption.

2.2. Sensors connection and circuit assembly

To connect the sensors and modules to the ESP32, each component is interfaced according to its specific communication method. The MICS-6814 gas sensor, which is able to detect gases like carbon monoxide (CO), nitrogen dioxide (NO₂), and ammonia (NH₃), has three analog output channels. These outputs are connected to the analog input pins of the ESP32, allowing continuous monitoring of various air pollutants. The BMP280 sensor, used to measure atmospheric pressure and temperature, communicates via the I2C protocol and is connected through the SDA and SCL pins, ensuring efficient and accurate data transfer. The GP2Y1010AU0F PM2.5 sensor detects particulate matter in the air and sends an analog voltage signal that is read by one of the ESP32's analog pins. This voltage varies depending on the particle concentration, enabling real-time air quality analysis.

2.3. Integration of sensors with LoRa Module and GPS module

To build a complete and connected air quality monitoring system, the LoRa module and NEO-6M Ublox GPS module are integrated with the ESP32 alongside the sensors. The LoRa module is connected using digital I/O pins, typically through SPI communication allowing the ESP32 to wirelessly transmit collected sensor data over long distances with minimal power usage. The NEO-6M GPS module connects to the ESP32 via UART, using the TX and RX pins to send real-time location data. This enables tagging of air quality readings with geographical coordinates, which is valuable for mapping pollution levels across different areas.

2.4. Development of a mobile application

A mobile application is developed by software team to provide a user-friendly interface for monitoring air quality data collected by the sensor system developed by hardware team. The app connects to the ESP32-based hardware either via internet using LoRa gateway. The app displays data in a clean and interactive dashboard, using color-coded indicators and live graphs to show trends over time. Users can view their current location on a map with a location marker based on GPS data, helping visualize pollution levels geographically. Additionally, the app can store historical data, offer alerts if pollution exceeds safe limits. The app is developed using Flutter. It serves as the user-end of the system, making complex environmental data easily accessible and understandable, and plays a crucial role in promoting public awareness and smart environmental decisions.

2.5. Integration of circuit with an application using Firebase

To enable real-time monitoring and data storage, the integration of the sensor circuit with a mobile application is achieved using Firebase, a cloud-based platform by Google. The ESP32 microcontroller, which gathers data from sensors like MICS-6814, BMP280, GP2Y1010AU0F, and the NEO-6M GPS module, is programmed to connect to Firebase using Wi-Fi. Through the Firebase Realtime Database, the ESP32 uploads sensor readings such as gas levels, temperature, pressure, PM2.5 concentration, and GPS coordinates in structured JSON format. On the application side, which is developed using Flutter, the integration with Firebase is seamless and highly efficient. Using Firebase SDKs, the app is able to fetch and display sensor data in real time. As soon as the ESP32 uploads new real-time readings to the Firebase Database, the app automatically receives and reflects these updates without delay. The user interface is designed to present this data clearly and intuitively through interactive dashboards, dynamic charts and real-time map views. This allows users to instantly understand current air quality levels, view trends, and monitor location-specific pollution data with ease. The combination of Flutter and Firebase provides a powerful platform for building responsive air quality monitoring system.

2.6. Testing and validation of the prototype

The sensor system was tested for sensor accuracy, data transmission, and overall functionality under various environmental conditions. Each component was evaluated to ensure reliable performance before real-world deployment. To achieve precise and dependable results, we calibrated the low-cost sensors using high-accuracy, certified reference-grade air quality monitoring instruments, ensuring their measurements closely align with standard benchmarks used in professional environmental monitoring.

Figure 1 represents schematic circuit diagram involved in circuit assembly which includes connection of sensors, LoRa module with an ESP-32 microcontroller.

Figure 2 represents schematic diagram of Air Quality Application workflow

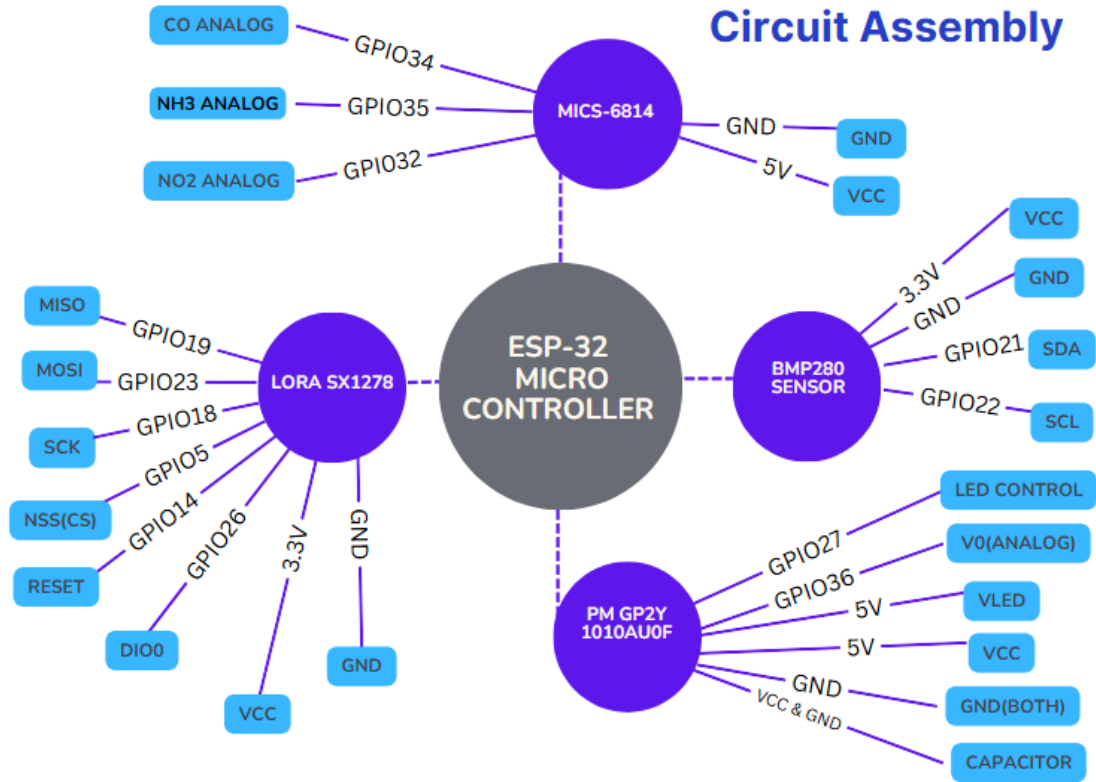


Figure 1: Schematic representation of circuit assembly and connections

3. Results and Discussions

This section gives finding from the implementation of Air-quality monitoring system with low cost-sensors. Observations from real-time data collection will be discussed in this section. [Section 3.1](#) deals with how well the integrated components are working. [Section 3.2](#) consists of analysis of data obtained using air-quality sensors about the real time updates and dashboard display of collected data. [Section 3.3](#) discusses about Sub-Indices calculations and AQI calculation. [Section 3.4](#) deals with advantages of integrated system. [Section 3.5](#) discusses about the challenges and limitations in the prototype developed.

3.1. System Functionality analysis

The circuit system integrated with sensors have demonstrated a responsive functionality in its test period. Data gathered by sensors was updated without any significant delay which clearly shows the efficient communication between ESP-32 micro-controller and low-cost sensors used in the circuit. The circuit assembly with sensors succeeded in capturing real-time variations in the data of air-quality parameters. Data Transmission to firebase cloud storage is also consistent in its functionality. These readings are also transferred to Database without any considerable delay. The application interface also responded and performed well in showing live data recorded by sensors. Changes in Air-Quality parameters were quickly reflected in a mobile application display. This ensured an effective

user experience in monitoring those air-quality parameters. The app also provided alerts when pollutant levels exceeded safe thresholds, enabling timely actions. Overall, the system promoted user engagement by making environmental data understandable, actionable, and accessible in everyday scenarios.

3.2. Data Collection and Analysis

In this Project, we collected air-quality using our sensor-based circuit-setup. The data was time-stamped and collected at regular intervals to help us analyze patterns over different times. These collected data at time intervals are sent to firebase and stored there based on time-stamps. Whenever we access data at a particular location, GPS module in a circuit record it's latitude and longitude and store that air-quality parameters data with respect to particular location along with time-stamp. The collected data is visualised in the form of graphs showing their trends with past-recorded values. The app also consists of a dashboard showing values of those parameters which were taken lastly. Figure 3 represents data recorded in an application developed, which are values taken continuously with regular intervals of 5 minutes. This is how typically our app shows data recorded by sensors in the form of charts.

3.3. AQI Calculation

Air Quality Index (AQI) is a numerical scale used to represent the quality of the air in a specific location. It simplifies complex data from multiple air pollutants into a single value

App Development

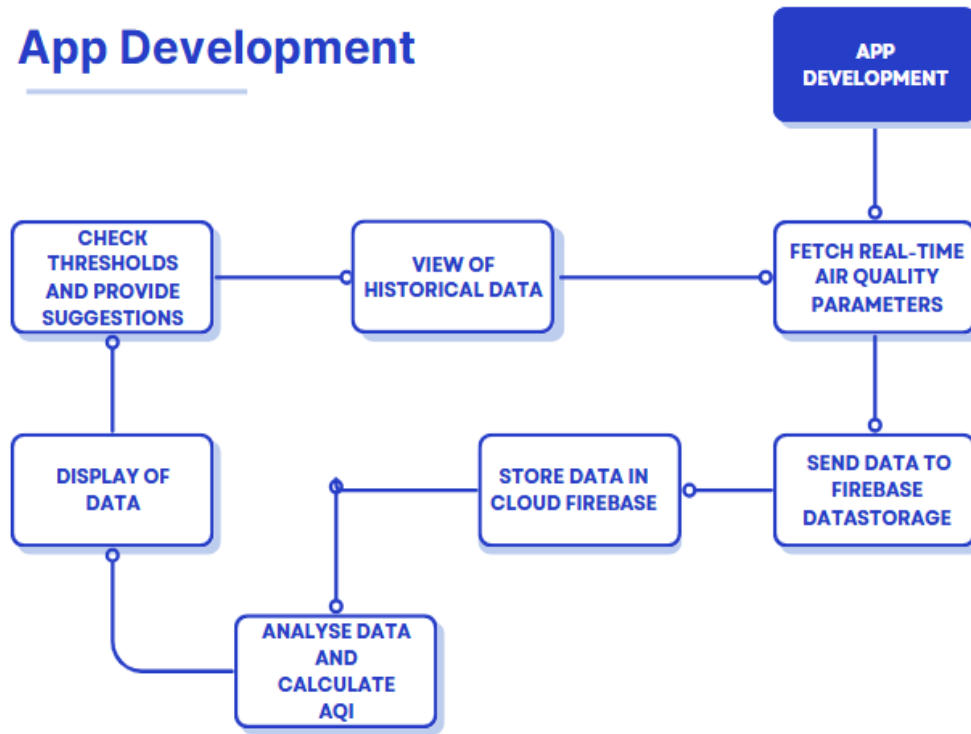


Figure 2: Schematic representation of Air Quality Application Workflow

that reflects how clean or polluted the air is, and what associated health effects might be expected. The AQI is calculated for individual pollutants called sub-AQIs. In our system, the following pollutants were measured using low-cost sensors:

- PM2.5 (Particulate Matter less than 2.5 microns)
- NO₂ (Nitrogen Dioxide)
- CO (Carbon Monoxide)

Each pollutant has predefined concentration breakpoints that correspond to AQI ranges, as published by the Central Pollution Control Board (CPCB), India. Once we calculate sub-indices of required air-quality parameters, we can calculate Air-Quality Index (AQI).

3.3.1. PM2.5 (24-hour average)

PM2.5 (µg/m ³)	AQI Range
0 – 30	0 – 50
31 – 60	51 – 100
61 – 90	101 – 200
91 – 120	201 – 300
121 – 250	301 – 400
251 – 500	401 – 500

These are standards for PM 2.5 concentration given by CPCB India.

3.3.2. NO₂ (1-hour average)

NO ₂ (µg/m ³)	AQI Range
0 – 40	0 – 50
41 – 80	51 – 100
81 – 180	101 – 200
181 – 280	201 – 300
281 – 400	301 – 400
401 – 800	401 – 500

These are standards for NO₂ concentration given by CPCB India.

3.3.3. CO (1-hour average)

CO (mg/m ³)	AQI Range
0 – 1	0 – 50
1.1 – 2	51 – 100
2.1 – 10	101 – 200
10.1 – 17	201 – 300
17.1 – 34	301 – 400
34.1 – 50	401 – 500

These are standards for CO concentration given by CPCB India.

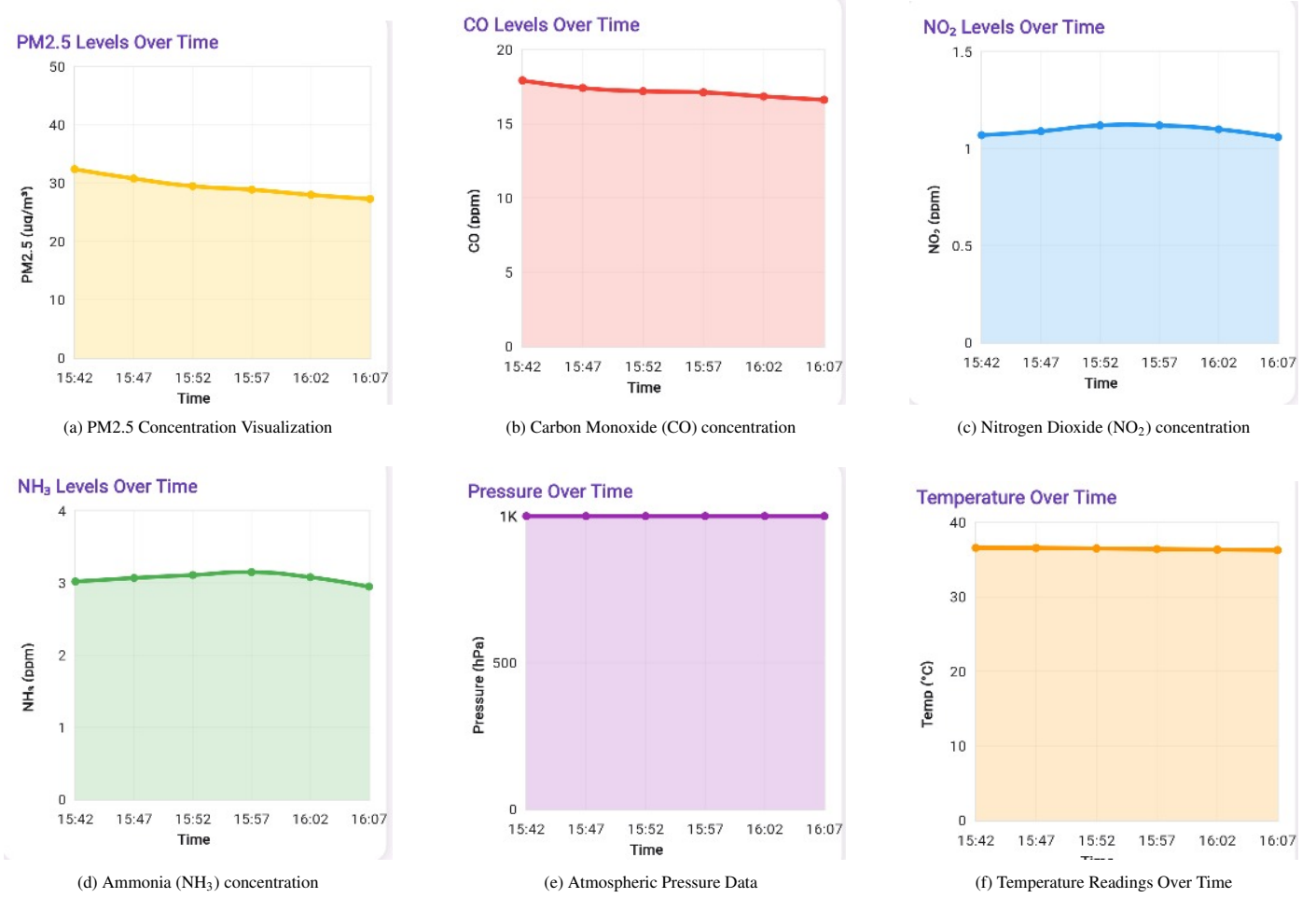


Figure 3: Visualizations of various air quality parameters captured by the sensor circuit system

To compute the AQI for a pollutant using the measured concentration, we apply the linear interpolation formula defined by CPCB:

$$I_p = \left(\frac{I_{hi} - I_{lo}}{BP_{hi} - BP_{lo}} \right) (C_p - BP_{lo}) + I_{lo}$$

Where:

- I_p : AQI for the pollutant
- C_p : Measured concentration of the pollutant
- BP_{hi}, BP_{lo} : Breakpoint concentrations that bracket C_p
- I_{hi}, I_{lo} : AQI values corresponding to the breakpoints

Finally, the overall AQI is computed as:

$$AQI = \max(AQI_{PM2.5}, AQI_{NO_2}, AQI_{CO})$$

We have performed measurement of these parameters on 19-04-2025, we have done as per CPCB India guidelines and obtained values are $PM2.5 = C_p = 23.5 \mu g/m^3$, $CO = C_p = 0.3415 \mu g/m^3$ and $NO_2 = C_p = 5.64 \mu g/m^3$.

3.3.4. PM2.5 AQI Calculation

The measured PM2.5 concentration be $C_p = 23.5 \mu g/m^3$. From the breakpoint table:

- $BP_{lo} = 0 \mu g/m^3, I_{lo} = 0$
- $BP_{hi} = 30 \mu g/m^3, I_{hi} = 50$

Applying the formula:

$$I_p = \left(\frac{50 - 0}{30 - 0} \right) (23.5 - 0) + 0$$

$$I_p = \left(\frac{50}{30} \right) (23.5) + 0$$

$$I_p \approx (1.667)(23.5) + 0$$

$$I_p \approx 39.174 + 0 = 39.174$$

Therefore, the AQI for PM2.5 at $23.5 \mu g/m^3$ is approximately:

$$\boxed{39.18}$$

3.3.5. CO AQI Calculation

The measured CO concentration be $C_p = 0.3415 \mu\text{g}/\text{m}^3$.
From the breakpoint table:

- $BP_{lo} = 0 \mu\text{g}/\text{m}^3, I_{lo} = 0$
- $BP_{hi} = 1 \mu\text{g}/\text{m}^3, I_{hi} = 50$

Applying the formula:

$$I_p = \left(\frac{50 - 0}{1 - 0} \right) (0.3415 - 0) + 0$$

$$I_p = \left(\frac{50}{1} \right) (0.3415) + 0$$

$$I_p \approx (50)(0.3415) + 0$$

$$I_p \approx 17.075 + 0 = 17.075$$

Therefore, the AQI for CO at $0.3415 \mu\text{g}/\text{m}^3$ is approximately:

$$\boxed{17.08}$$

3.3.6. NO₂ AQI Calculation

The measured NO₂ concentration be $C_p = 10.35 \mu\text{g}/\text{m}^3$.
From the breakpoint table:

- $BP_{lo} = 0 \mu\text{g}/\text{m}^3, I_{lo} = 0$
- $BP_{hi} = 40 \mu\text{g}/\text{m}^3, I_{hi} = 50$

Applying the formula:

$$I_p = \left(\frac{50 - 0}{40 - 0} \right) (10.35 - 0) + 0$$

$$I_p = \left(\frac{5}{4} \right) (10.35) + 0$$

$$I_p \approx (1.25)(10.35) + 0$$

$$I_p \approx 12.9375 + 0 = 12.9375$$

Therefore, the AQI for NO₂ at $10.35 \mu\text{g}/\text{m}^3$ is approximately:

$$\boxed{12.94}$$

Finally, the overall AQI is computed as:

$$\text{AQI} = \max(\text{AQI}_{\text{PM}_{2.5}}, \text{AQI}_{\text{NO}_2}, \text{AQI}_{\text{CO}})$$

$$\text{AQI} = 39.18 \text{ — Good AQI category}$$

3.4. Advantages and Limitations of sensor system

The present project is cost-effective and scalable deployment. It empowers users with localized air quality insights and supports early intervention through suggestions. The mobile application is designed with a interactive dashboard that allows users to monitor real-time pollutant levels (PM_{2.5}, CO, and NO₂) with minimal effort. The app's interface has been tested for responsiveness, ensuring a smooth experience, since the system runs in the background and updates automatically. The mobile app integrates seamlessly with real-time sensor data, displaying pollutant concentrations and AQI values instantly, making environmental monitoring accessible to everyone. Main disadvantage of our sensor system, we did not include sensors which can measure SO₂, O₃ and PM₁₀ because incorporating these would have significantly increased the cost and complexity of the system. Moreover, sensors for these pollutants often require regular calibration and maintenance to ensure accuracy, which can be challenging in low-resource or unattended deployments.

4. Conclusion

The project aimed to design and implement a low-cost, smart air quality monitoring system, and it has successfully achieved its core objectives. The integration of sensors capable of detecting PM_{2.5}, CO, and NO₂ with an ESP32 microcontroller allowed for real-time air quality data collection. This data was transmitted seamlessly to Firebase, enabling users to view live readings through a mobile application. The system maintained reliable performance across normal environmental conditions, and users were able to interact with the app intuitively without any technical barriers.

One of the major strengths of the system is its mobile-friendly design, which provides a simple and effective way for users to stay informed about air pollution levels. The automatic updates, alert features, and easy-to-read data presentation make it suitable for daily use. While occasional sensor fluctuations were noticed in humid environments, the overall results support the system's usefulness for continuous, location-based monitoring.

In conclusion, this system proves to be a practical and scalable solution for promoting environmental awareness. With further tuning and improvements, especially in calibration and weather adaptation, the system can be adopted in larger networks for smart city applications, schools, residential areas, or any location needing air quality insights in real time.

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[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15]
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